Probability :-

Probability tell us how likely an event is to happen.

$$\text{Probability} = \frac{\text{No. of favorable outcomes}}{\text{Total possible outcomes}}$$

Ex:- Coin toss $\Rightarrow \{H, T\} \Rightarrow \boxed{P(H) \Rightarrow 1/2}$

Head will occur 50% of the time

Random Experiment:-

→ A random Experiment is an action or process
- than can be repeated
- has multiple possible outcomes
- But the Exact outcome is unpredictable.

→ we know all possible outcomes, but not which one will occur.

Ex:- tossing a coing, rolling a die.

Trial:- a trial is one single performance of random Exper

⇒ multiple trials → help to Estimate probabilities

Outcome:- Outcome refers to a single possible result of trial

Sample Space:- The sample space is the set of all possible outcomes of a random Experiment.

$S = \{$ all possible outcomes $\}$

Ex:- Coin toss $\Rightarrow S = \{H, T\}$, dice roll, $S = \{1, 2, 3, 4, 5, 6\}$

ML
$S = \{$Spam, not Spam$\}$

Event:- An Event is any Subset of the sample space. It contains one outcome or multiple outcome

Ex:- dice, $S = \{1, 2, 3, 4, 5, 6\}$,

events

i) getting even number $S = \{2, 4, 6\}$

ii) getting six number $S = \{6\}$

Ex:- R.E ⇒ Tossing a coin twice

Trial ⇒ Tossing coin twice (once)

Outcome ⇒ $\{H, T\}$

S.S ⇒ $\{(H,H), (HT) (TH), (TT)\}$

Event ⇒ i) getting 2 (H,H) ⇒ $\{H, H\}$

ii) getting atleast head $\{HH, HT, TH\}$

ML Ex:- Titanic ⇒ 891 passengers ⇒ Pclass

R.E = Randomly picking / drawing out a passenger

trial ⇒ finding their Pclass

outcomes ⇒ P class 1 {1}

SS ⇒ {1, 2, 3}

event ⇒ passenger is from P class2 ⇒ {2}

passenger is not from Pclass = 2 ⇒ {1, 3}

Types of Event:-

i) Simple Event:- A simple event containly only one outcome

Ex:- Event (3) ⇒ 1/1 $S = \{1, 2, 3, 4, 5, 6\}$

→ only 1 possible outcome

ii) Compound Event:- A Compound event contains more than one outcom

Ex:- i) getting even no., ⇒ $B = \{2, 4, 6\}$

iii) Certain / Sure Event:- Event that always occurs

Ex:- Getting 1 ⇒ {1} , getting 6 ⇒ {6} = $\boxed{1 \le x \le 6}$

iv) Impossible event:- An Event that can never occur

Ex:- getting no 7 ⇒ {~~6~~} ⇒ impossible

iv) Mutually Exclusive Events:- Two Events that cannot occur at the same time.

Ex:- getting 2 & 5 in one roll ⇒ not possible

Ex:- Spam vs Not spam are mutually Exclusive classes.

v) Non mutually Exclusive:- Events that can occur together

Ex:- getting an Even no. ⇒ $A = \{2, 4, 6\}$

getting an odd no ⇒ $B = \{1, 3, 5\}$

vi) Exhaustive Events:-

A set of Events that covers the sample space

Ex:- Event A = {1, 3, 5} ⇒ all possibles covered Sample

event B = even ⇒ {2, 4, 6} Space ⇒ {1, 2, 3, 4, 5, 6}

Theoritical vs Empirical Probability :-

Theoritical Probability :-

Theoritical Probability is calculated before performing the Experiment, assuming all comes are equally likely.

$$\boxed{P(A) = \frac{\text{Number of Favorable outcomes}}{\text{Total possible outcomes}}}$$

rule
Equaly likely

→ here using any previous data

→ Ex: Rolling a die ⇒ $S = \{1, 2, 3, 4, 5, 6\}$

$P(1) = P(2) = P(3) = P(4) = P(5) \Rightarrow P(6) \Rightarrow \boxed{1/6}$

→ (here assuming all outcomes are Equally likely) assumption

→ we didn't roll the die ⇒ we used logic & symmetry

Emperical Probability :- [observed / Experimental]

Emporical Probability is calculated after performing the Experiment using actual observed data.

$$\boxed{P(A) \Rightarrow \frac{\text{No. of Events Occurred}}{\text{Total No. of Trial}}}$$

Ex:- Coin Toss (100 times) → 100 trials

- Head appeared = 60 times } event occurred $\boxed{\Rightarrow P(A) = \frac{60}{100} = 0.6}$
- Total trials ⇒ 100 times

→ based on real observation, not assumptions.

when ⇒ $\boxed{\text{No. of trials}}$ ⇒ $\boxed{\text{infinite Coin toss}}$

⇓

Empirical prob ⇒ Comes close to Theoritical Prob

$\boxed{P(H) = \frac{1}{2}} \Rightarrow \boxed{0.5}$

	Consider head	
10 ⇒	3	⇒ 0.3
100 ⇒	45	⇒ 0.45
1000 ⇒	470	⇒ 0.47

→ Machine Learning models rely on Empirical Probability, not theorical. Ex:- Spam Classification

word	Occurs in Spam	Total Spam emails
free	800	1000

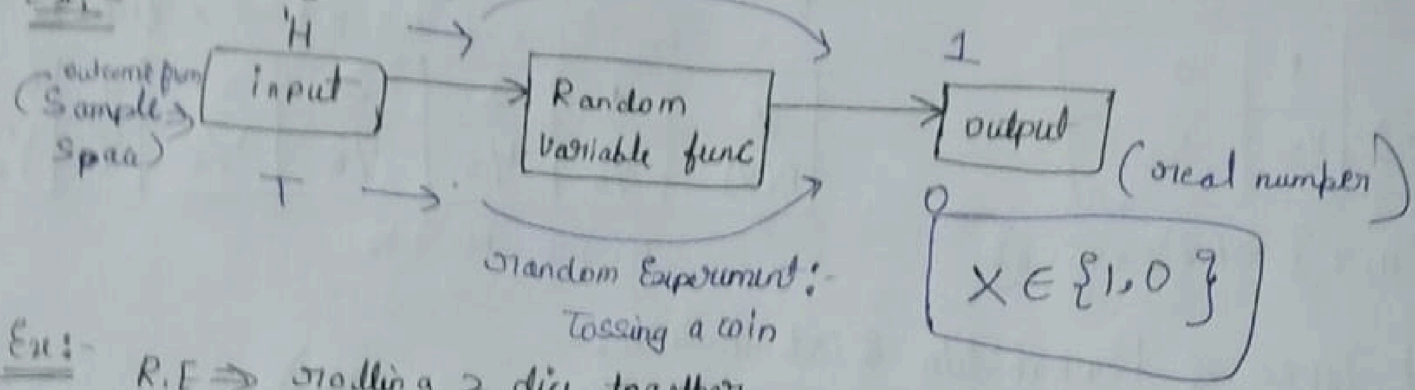
$\boxed{P(\text{free} \mid \text{spam}) \Rightarrow 800/1000}$

Wt is Random Variable :-

→ A Random Variable is a function that assigns numerical Value to each outcome of a random Experiment.

- It converts outcomes into numbers, So we can do math on the

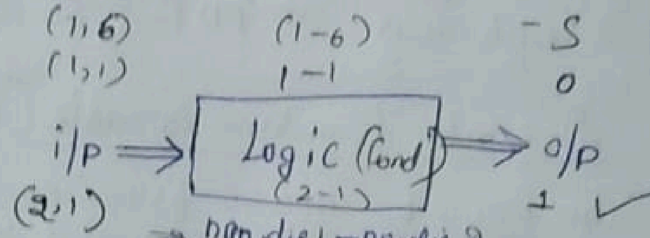
Ex:- $S = \{H, T\}$



Ex:- R.E ⇒ rolling 2 dice together

Event A ⇒ where the num of die 1 > where no. of die 2

SS = { (1,1), (1,2), (1,3), (1,4) (1,5) (1,6)
(2,1)
⋮
(6,1) (6,2) (6,3) (6,4) (6,5) (6,6) }



⇒ num die1 − no. die 2
if die1 > die2, return T
elif die1 < die2, return F
else die1 == die2

Types of Random Variable :-

1. Discrete Random Variable :-

- Takes countable values, usually integers

Ex:- no. of heads in 2 coin tosses
$X = \{0, 1, 2\}$

Continuous Random Variable :-

- Takes any value in an interval (range)
- infinite possible values

Ex:- heigh of a student ⇒ outcome: Exact height (165.7cm)
$X =$ height in cm, so $X \in R$, real num

∴ continuous random variable map to real numbers

why real number

→ math Operations (mean, Variance) need numbers
→ Probability distributions are defined over numbers
→ ML models works only with numeric values

Probability Distribution of a Random Variable :-

A Probability Distribution is a list of all the possible outcomes of a random variable along with their corresponding Probability values

i) Coin Toss

X	1	0
P(X)	1/2	1/2

$\{H, T\} \Rightarrow RV \Rightarrow x = \{1, 0\}$

ii) rolling a die

$SS = \{1, 2, 3, 4, 5, 6\} \Rightarrow X = \{1, 2, 3, 4, 5, 6\}$

X	1	2	3	4	5	6
P(X)	1/6	1/6	1/6	1/6	1/6	1/6

$P(X=1) = 1/6$ $P(X=2) = 1/6$ ......

Types of Probability Distribution :-

→ Discrete ⇒ PMF (Probability mass function)

→ Continuous ⇒ PDF (Probability Density function)

Descrete Random Variable (PMF) :-

The Probab mass function gives $P(X = x)$, for each value of x

$$\boxed{\Sigma P(X = x) = 1}$$

X	P(X)
0	1/2
1	1/2

⇒ $X = 1$ if head

⇒ $X = 0$ if tail

Continuous Random Variable :- (PDF)

The Probability Density function gives density of probability ⇒ $f(x)$

→ $P(X = x) = 0$

→ Probabilities are calculated over intervals

Ex :- $\boxed{X = \text{height (cm)}}$ $\boxed{\text{Probability} = P(160 \leq x \leq 170) = \int_{160}^{170} f(x)\, dx}$

Mean of a Random Variable :-

The mean of a Random Variable, often called expected value is essentially the average outcome of a random process that is repeated many times.

→ more technically, its weighted average of the possible outcomes of the random variable, where each outcome is weighted by its probability of occurance

Ex:- rolling a die

X	1	2	3	4	5	6
P(X)	1/6	1/6	1/6	1/6	1/6	1/6

$\Rightarrow$

$\boxed{\text{mean of } X}$

$\Downarrow$

multiple trials

$\Downarrow$

1000 die roll

$\Downarrow$

$$\frac{3+4+2+5+6+\dots+7}{1000} \Rightarrow \text{mean}$$

i) Ex:- 5 3 4 5 3 3

$$\Rightarrow \frac{5+3+4+5+3+3}{6} \Rightarrow \frac{23}{6}$$

ii) $$\frac{2(5)+1(4)+3(3)}{6} \Rightarrow \frac{2}{6}(5)+\frac{1}{6}(4)+\frac{3}{6}(3)$$

$$\Rightarrow \frac{1}{3}(5)+\frac{1}{6}(4)+\frac{1}{2}(3)$$

$$\boxed{E[X] = \sum_{i=1}^{n} X_i P(X_i)}$$

Variance of a random Variable:-

Variance measures how much the values of a random variable spread out from the mean.

$$\boxed{\text{Variance} = Var(X)}$$

- mean tells center
- Variance tells spread

$\rightarrow$ The variance of a random variable is a statistical measurement describes how much individual observations in a group differ from the mean (Expected value).

Normal Variance $\Rightarrow$ $\boxed{\sum_{i=1}^{n} \frac{(X_i - \bar{X})^2}{n-1}}$ $\Rightarrow$ (2, 4, 5, 6, 7 .... 10)

Variance $\Rightarrow$ spread $\} \Rightarrow$ avg

$\rightarrow$ for rolling a die

$X \Rightarrow 1\ 2\ 3\ 4\ 5\ 6 \}\overset{avg}{=3.5}$ $\Rightarrow (X - E[X])^2$

$\Downarrow$

$(1-3.5)^2 + (2-3.5)^2 +$

$(3-3.5)^2$ ..... $(6-3.5)^2$

$\Downarrow$

of mean $\Rightarrow$ Expected value

$\Downarrow$

$$\boxed{Var(X) = E\left[(X - E[X])^2\right]}$$

or

$$\boxed{Var(X) = E[X^2] - (E[X])^2}$$

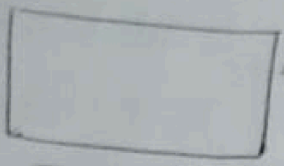
from formula

Ex:- X = 1 (head), X = 0 (Tail)

mean $\Rightarrow E[X] = 0.5$

$Var(X) \Rightarrow (1-0.5)^2(0.5) + (0-0.5)^2(0.5) \Rightarrow 0.25$

Venn Diagrams in Probability :-

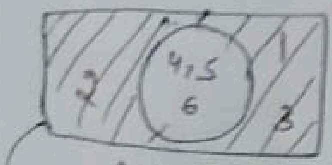


$\Rightarrow U$ {sample space}

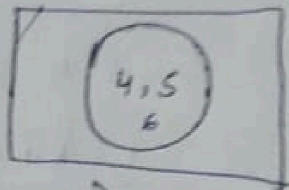
$\boxed{P(U) = 1}$

Ex: rolling a die

i) Event $A \rightarrow$ getting num ≥ 4



$P(A') = P(U) - P(A)$
$= 1/2$

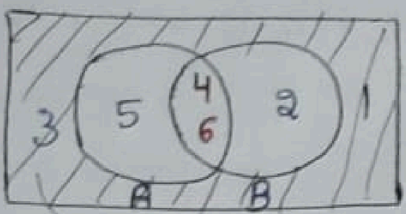


$P(A) = 3/6 = 1/2$

Ex 2:-

Event $A \rightarrow$ getting num ≥ 4 $\Rightarrow$ S.S $\Rightarrow \{4, 5, 6\}$

Event $B \rightarrow$ even num $\Rightarrow$ S.S $\Rightarrow \{2, 4, 6\}$



$\rightarrow P(A \cap B) \Rightarrow 2/6 \Rightarrow 1/3$ $\{2, 6\}$

$\rightarrow P(A) \Rightarrow 3/6 \Rightarrow 1/2$, $P(B) \Rightarrow 1/2$

$\rightarrow P(A \cup B) = \cancel{4}^{2}/\cancel{6}_{3} \Rightarrow \{2, 4, 5, 6\}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 1/2 + 1/2 - 1/3$
$= \frac{3+3-2}{6} \Rightarrow 4/6$

$\rightarrow P(A \cup B)' = 2/6 =$
$P(U) - P(A \cup B)$
$1 - 2/3 \Rightarrow 1/3$

Contigency Table in Probability :-

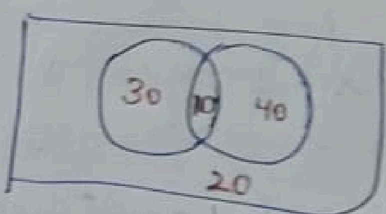
$\rightarrow$ Contigency Table Shows :
$\Rightarrow$ How events occur together $\Rightarrow$ The relationship between Variables

Ex:- Email classification :- Suppose we observe 100 emails

	Spam	Not Spam	Total
word = free	30	10	40
word ≠ free	20	40	60
Total	50	50	100

Ex 2:- 100 students $\Rightarrow$ 40 $\rightarrow$ only bio 30 $\rightarrow$ only math , 10 $\rightarrow$ both

	math	not math	total
bio	10	40	50
not bio	30	20	50
	40	60	100



Types of Probabilities:-

i) Joint Probability:-

Joint Probability is the probability that two events occur together.

→ Lets say we have two random variable X and Y. The Joint probability of X and Y, denoted as $P(X = x, Y = y)$, is the probability that X takes the value x and y takes value y at the same time

Ex:- X → Random variable associated with Pclass of a passenger
Y → " — " — " with survival status

Pclass	1	2	3
Survived			
0	80	97	372
1	136	87	119

Joint prob ⇒ total passenger = 891

P class	1	2	3
Survived	~~0.089~~	~~0~~	
0	0.089	0.108	0.417
1	0.015	0.097	0.133

⇒ 1

→ Each cell / total observation ⇒ Joint Probability

→ Join Probability distribution

$\rightarrow P(\overset{X}{\text{Pclass } 1}, \overset{Y}{\text{Survived } 0})$
$\Rightarrow 80/891 \Rightarrow 0.089$

$\rightarrow P(\overset{X}{\text{Pclass } 3}, \overset{Y}{\text{Survived } 1})$
$372/891 \Rightarrow 0.133$

⇒ $\boxed{P(A \cap B)}$

Ex) $P(\text{Pclass}=1) \Rightarrow 0.24$
$P(\text{Survived}=0) \Rightarrow 0.616$

ii) Marginal Probability:- [Simple][unconditional]

marginal Probability is the probability of one event, obtained by summing over the other variable.

→ It's called marginal because we get it from the margin (total)

Pclass	1	2	3	
Survived				
0	80	97	372	549
1	136	87	119	342
Total	216	184	491	891

Pclass	1	2	3	
Survived				
0	0.089	0.108	0.417	0.616
1	0.015	0.097	0.133	0.38
Total				

Conditional Probability :-

Conditional Probability is the probability of an Event A, given that another event B has already occured.

$\boxed{P(A|B)}$ ⇒ read as: Probability of A given B

→ Once B is knows, the sample space shrinks to only those outcomes where B occured.

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

S.S ⇒ {HHH, HHT, HTH, THH, ~~TTT~~, TTH, THT, HTT} ⇒ 8

B ⇒ Question: Three unbaised coins are tossed, what is cond Prob, that at least two coins shows head, given that at least one coint show heads?

A ⇒ at least two coins head (≥ 2 heads)

B ⇒ at least 1 head (≥ 1 Head) [T,TT] canceld / shrunked from 8 to 7

So ⇒ sample Space = 7

B ⇒ 7, A = 4

$$\boxed{P(A|B) = \frac{4}{7}}$$

Question2:- 2 fair six-sided dice are rolled. What is the cond Probability that the sum of the no rolled is 7, given that first die shows an odd number

Total = 36

A ⇒ Sum of die = 7 (6)

B ⇒ first dies is odd (1, 3, 5)th row ⇒ (18) cancelled

S.S = {~~(1,1) (1,2) (1,3) (1,4) (1,5) (1,6)~~
(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)
~~(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)~~
(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)
~~(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)~~
(6,1) (6,2) (6,3) (6,4) (6,5) (6,6)}

only cancelled we considerize

$$\boxed{P(A|B) = 3/_{18} = \frac{1}{6}}$$

question 3:- Conditional Prob ~~that~~

Two fair dice are rolled, denoted as D1 and D2, what is the C.P that D1 Equals D2. given that the sum of D1 & D2 is less than or Equal 5.

{ $\boxed{(1,1)}$ $\boxed{(1,2)}$ $\boxed{(1,3)}$ $\boxed{(1,4)}$ (1,5) (1,6)
$\boxed{(2,1)}$ $\boxed{(2,2)}$ $\boxed{(2,3)}$ (2,4) (2,5) (2,6)
$\boxed{(3,1)}$ $\boxed{(3,2)}$ (3,3) (3,4) (3,5) (3,6)
$\boxed{(4,1)}$ (4,2) (4,3) (4,4) (4,5) (4,6)
(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)
(6,1) (6,2) (6,3) (6,4) (6,5) (6,6) }

A → D1 ⇒ D2

10 (B ⇒ D1 + D2 ≤ 5
26 cancelled)

$$\boxed{P(A|B) \Rightarrow \frac{3}{10}}$$

formula:-

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

P(A∩B) → Joint probability of A & B

P(B) → marginal Probability of

Intuition behind Conditional Probability:-

→ C.P works by changing the Sample Space. when we write P(A|B) we are assuming that event B has already happend. So outcom where B did not occur are no longer relevant. This means B itself becomes the new sample space.

→ P(B) represent the size of this new sample sample space, so we divide by it to normalize probabilities

→ P(A∩B), represents the outcomes where both A & B occur together

→ dividing P(A∩B) by P(B) gives the fraction of B-outcomes in which A occurs.

→ So, P(A|B) tells us how likely A is when B is already known to have happened.

Independent VS Mutually Exclusive Events :-

→ **Independent Events :-** are Events where the occurence of one event does not affect the occurence of another

Ex:- 1. flipping a coin & rolling die
2. drawing a card with replacement

→ **Dependent Events :-** occurence of one even does affect the occurance of another

Exs- 1. drawing a card with replacement ⇒ 4/51

	if statistical independent
$P(A\|B)$ =	$P(A)$
$P(B\|A)$ =	$P(B)$
$P(A \cap B)$ =	$P(A)\,P(B)$

$$P(A|B) \Rightarrow \frac{P(A\cap B)}{P(B)} \Rightarrow \frac{P(A)\cancel{P(B)}}{\cancel{P(B)}}$$

Mutually Exclusive Events:

Events that cannot occur at the same time, meaning if one event occurs, the othe cannot.

Ex:- flipping a coin $\{H, T\}$ ⇒ can't occur at same time
rollin a a dice ⇒ $P(A \cap B) = 0$

$P(A|B) = 0$ $\boxed{P(A \cap B) = 0}$

$P(B|A) = 0$

Bayes Theorem:-

→ Bayes Theorem is a fundamental concept in the field of probability and statistics that describes how to update the probability of hypothesis when given evidence

→ It is used in a variety of fields, including machine learning, statistics and game theory.

$$\boxed{P(A|B)} = \boxed{P(A)} \times \frac{\boxed{P(B|A)}}{\boxed{P(B)}}$$

likelihood { how likely B is when A is true

Posterior (updated belief about A after seeing B)

Prior (what we believe about A before seeing B)

marginal / Evidence (normalizing)

mathematical Proof:-

from Conditional Probability $\Rightarrow P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} \Rightarrow P(B \cap A) = P(A \cap B) = \underline{P(B|A)P(A)}$$

$$\boxed{P(A|B) = \frac{P(A)P(B|A)}{P(B)}}$$

Ex:-

gender	survived
M	0
M	0
F	1
F	0
M	1

$\}2/3$

$$\underline{P(0|M)} \Rightarrow \frac{P(M|0)\,P(0)}{P(M)}$$

marginal

$$\Rightarrow \frac{2/3 \times 3/\cancel{5}}{3/\cancel{5}}$$

$$\Rightarrow 2/3$$

$$\underline{P(1|M)} \Rightarrow \frac{P(M|1)\,P(1)}{P(M)}$$

$$\Rightarrow \frac{1/2 \times 2/5}{3/5} = 1/3$$

$$\boxed{P(1|m) > P(0|m)}$$

So, probability of ~~dying~~ of gender male is more than survival.

→ Bayes Theorem provides a way to update probability of an event based on a new Evidence by Combining prior knowledge with the observed data